

Stroke Prediction Using Machine Learning

Project Title

Stroke Prediction Using Gradient Boosting Algorithm

Objective

The objective of this project is to predict whether a person is likely to have a stroke based on healthcare-related attributes using Machine Learning. The model helps identify individuals at higher risk by analyzing patient information.

Dataset

Dataset Name: Healthcare Stroke Dataset

Features Used

- Gender
- Age
- Hypertension
- Heart Disease
- Ever Married
- Work Type
- Residence Type
- Average Glucose Level
- BMI
- Smoking Status

Target Variable:

- Stroke (0 = No Stroke, 1 = Stroke)
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Data Preprocessing

The following preprocessing steps were performed:

- Imported the dataset using Pandas.
- Removed the unnecessary **id** column.
- Checked for missing values.
- Filled missing BMI values using the mean value.
- Removed duplicate records.

- Checked the dataset information and summary statistics.
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Feature Engineering

Three new columns were created to improve model performance.

1. BMI Category

BMI values were categorized into:

- Underweight
- Normal
- Overweight
- Obese

2. Age Group

Age was divided into:

- Child
- Young Adult
- Adult
- Senior

3. Risk Score

A Risk Score column was created using:

- Hypertension
 - Heart Disease
 - High Glucose Level (>140)
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Data Encoding

Categorical columns were converted into numerical values using **Label Encoder**.

Encoded columns include:

- Gender
- Ever Married
- Work Type
- Residence Type
- Smoking Status

- BMI Category
 - Age Group
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Feature Scaling

The dataset was standardized using **StandardScaler** so that all features have the same scale before training the model.

Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

The split was performed using `train_test_split()` with a random state for reproducibility.

Machine Learning Algorithm

Initially, the K-Nearest Neighbors (KNN) algorithm was implemented.

Later, the model was improved by using the **Gradient Boosting Classifier**, which provided better prediction performance.

Model Evaluation

The model was evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report

The Gradient Boosting model achieved an accuracy greater than **80%**, making it suitable for stroke prediction.

User Prediction

A simple prediction interface was created where the user enters only:

- Age
- Average Glucose Level
- BMI

The remaining feature values are assigned default values, and the trained Gradient Boosting model predicts whether the person is likely to have a stroke.

Technologies Used

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Scikit-learn
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Conclusion

This project demonstrates how Machine Learning can be applied to predict stroke risk using healthcare data. After preprocessing, feature engineering, encoding, scaling, and model training, the Gradient Boosting algorithm produced reliable prediction results. The final system allows users to enter basic health information and receive a stroke prediction, making the model simple, efficient, and easy to use.